KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5492 | 4 Credits: 4 | Program Core | Source-grounded

Microcontroller and

● To equip students on architecture, interfacing and Assembly Language

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5492 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5492.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Microelectronics
Course code	5492
Course title	Microcontroller and
Semester	4 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● To equip students on architecture, interfacing and Assembly Language
- Programming (ALP) of 8051 microcontrollers.
- ● To enable students to develop simple interfacing applications for daily use
- ● To prepare students for troubleshooting of microcontroller-based equipment.

Course outcomes

CO	Official outcome	Study level
CO1	12 Understanding Microcontroller. Develop simple Assembly Language Programs	See official syllabus
CO2	17 Applying (ALP) for 8051. Explain interrupts, timer and serial communication	See official syllabus
CO3	14 Understanding in 8051 Illustrate the interfacing of various I/O devices with	See official syllabus
CO4	15 Understanding	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 3
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain the features and block diagram of 8051 Microcontroller.	4	As specified
Module 2	Develop simple Assembly Language Programs (ALP) for 8051.	4	As specified
Module 3	Explain interrupts, timer and serial communication in 8051.	4	As specified
Module 4	Illustrate the interfacing of various I/O devices with 8051	4	As specified

MODULE 1

Explain the features and block diagram of 8051 Microcontroller.

This module is presented from the official syllabus scope for Microcontroller and. Use the topic cards for learning and the source transcript for exact coverage.

As specified official hours	**CO-linked** see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the features and block diagram of 8051 Microcontroller.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Compare Microprocessor and Microcontroller

Compare Microprocessor and Microcontroller is an official topic in Module 1 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare Microprocessor and Microcontroller using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare microprocessor and microcontroller should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compare Microprocessor and Microcontroller means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Compare Microprocessor and Microcontroller □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051

List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051 is an official topic in Module 1 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list the features of 8051 microcontroller. 1 remembering explain the block diagram of 8051 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051 □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ 2 Understanding microcontroller.

2 Understanding microcontroller. is an official topic in Module 1 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding microcontroller. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding microcontroller. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding microcontroller. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 2 Understanding microcontroller. □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▼ **Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register.**

Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register. is an official topic in Module 1 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the memory of 8051. 7 understanding features and block diagram of 8051 introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, data memory organization, program memory organization, psw register. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Develop simple Assembly Language Programs (ALP) for 8051.

This module is presented from the official syllabus scope for Microcontroller and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Develop simple Assembly Language Programs (ALP) for 8051.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding Programming. Explain various addressing modes and types

3 Understanding Programming. Explain various addressing modes and types is an official topic in Module 2 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Programming. Explain various addressing modes and types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding programming. explain various addressing modes and types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding Programming. Explain various addressing modes and types means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Understanding Programming. Explain various addressing modes and types □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□.

▼ 4 Understanding of instructions. Describe Jump, Loop, Call and single bit level

4 Understanding of instructions. Describe Jump, Loop, Call and single bit level is an official topic in Module 2 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding of instructions. Describe Jump, Loop, Call and single bit level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding of instructions. describe jump, loop, call and single bit level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Understanding of instructions. Describe Jump, Loop, Call and single bit level means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding of instructions. Describe Jump, Loop, Call and single bit level □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▾ 4 Understanding instructions. Develop Assembly language programs for

4 Understanding instructions. Develop Assembly language programs for is an official topic in Module 2 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding instructions. Develop Assembly language programs for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding instructions. develop assembly language programs for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Understanding instructions. Develop Assembly language programs for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding instructions. Develop Assembly language programs for □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 6 Applying 8051. Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing.

6 Applying 8051. Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing. is an official topic in Module 2 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Applying 8051. Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 applying 8051. introduction to assembly language programming of 8051 – structure, addressing modes, types of instructions: -data transfer, arithmetic, logical, compare, rotate and swap instructions. describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, delay generation. single-bit level instructions, data serialization. simple programs: - 8-bit addition, 8-bit subtraction, multiplication, division, ascii to packed bcd, packed bcd to ascii, bcd to hex and hex to bcd, port reading and writing. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Applying 8051. Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 6 Applying 8051. Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain interrupts, timer and serial communication in 8051.

This module is presented from the official syllabus scope for Microcontroller and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain interrupts, timer and serial communication in 8051.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▾ 4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating

4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating is an official topic in Module 3 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding interrupts and handling of interrupts. explain tmod register and timer operating should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ 4 Understanding modes. Illustrate TCON register and time delay

4 Understanding modes. Illustrate TCON register and time delay is an official topic in Module 3 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding modes. Illustrate TCON register and time delay using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding modes. illustrate tcon register and time delay should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding modes. Illustrate TCON register and time delay means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding modes. Illustrate TCON register and time delay □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 3 Understanding calculation for Timer. Illustrate special function registers for serial

3 Understanding calculation for Timer. Illustrate special function registers for serial is an official topic in Module 3 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding calculation for Timer. Illustrate special function registers for serial using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding calculation for timer. illustrate special function registers for serial should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding calculation for Timer. Illustrate special function registers for serial means in the official course context
Study lens	Principle → components or stages → result → application
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Malayalam support: Module 3-□□ 3 Understanding calculation for Timer. Illustrate special function registers for serial □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 3 Understanding communication. Interrupts, Timers, Serial communication. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Timer: TMOD register, Timer operating modes, TCON register, Time Delay calculation. Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF register, SCON register and Different serial data transmission modes in 8051. PCON register.

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serial data transmission modes in 8051. PCON register. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding communication. interrupts, timers, serial communication. 8051 interrupts: concept, types, interrupt service routine (isr), ie register, steps in executing an interrupt, interrupt priority, ip register. timer: tmod register, timer operating modes, tcon register, time delay calculation. basics of serial communication, baud rate, 8051 connections to rs232 connector, sbuf register, scon register and different serial data transmission modes in 8051. pcon register. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding communication. Interrupts, Timers, Serial communication. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Timer: TMOD register, Timer operating modes, TCON register, Time Delay calculation. Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF register, SCON register and Different serial data transmission modes in 8051. PCON register. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding communication. Interrupts, Timers, Serial communication. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Timer: TMOD register, Timer operating modes, TCON register, Time Delay calculation. Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF register, SCON register and Different serial data transmission modes in 8051. PCON register. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Illustrate the interfacing of various I/O devices with 8051

This module is presented from the official syllabus scope for Microcontroller and. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the interfacing of various I/O devices with 8051
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 4 Understanding 8051. Explain the interfacing of DC motor for speed

4 Understanding 8051. Explain the interfacing of DC motor for speed is an official topic in Module 4 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding 8051. Explain the interfacing of DC motor for speed using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding 8051. explain the interfacing of dc motor for speed should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Understanding 8051. Explain the interfacing of DC motor for speed means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Understanding 8051. Explain the interfacing of DC motor for speed □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 3 Understanding control. Illustrate the interfacing of 4x4 keyboard and

3 Understanding control. Illustrate the interfacing of 4x4 keyboard and is an official topic in Module 4 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding control. Illustrate the interfacing of 4x4 keyboard and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding control. illustrate the interfacing of 4x4 keyboard and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding control. Illustrate the interfacing of 4x4 keyboard and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 4-□□ 3 Understanding control. Illustrate the interfacing of 4x4 keyboard and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▾ 4 Understanding 16x2 LCD system with 8051. Illustrate ADC, DAC and LM35 temperature

4 Understanding 16x2 LCD system with 8051. Illustrate ADC, DAC and LM35 temperature is an official topic in Module 4 of Microcontroller and. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding 16x2 LCD system with 8051. Illustrate ADC, DAC and LM35 temperature using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding 16x2 lcd system with 8051. illustrate adc, dac and lm35 temperature should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Understanding 16x2 LCD system with 8051. Illustrate ADC, DAC and LM35 temperature means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

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Malayalam support: Module 4-[illegible] 4 Understanding 16x2 LCD system with 8051. Illustrate ADC, DAC and LM35 temperature [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 4 Understanding sensor interfacing with 8051. Stepper motor – operation, explain the interfacing with 8051 DC motor – operation, explain the interfacing with 8051 4x4 -Matrix keyboard – Structure, identifying key press, explain the interfacing with 8051. 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051. Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor. Text / Reference: T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller T1 and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Subratha Ghoshal – 8051 microcontroller internals, instructions, programming T2 and interfacing- Pearson T3 Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition. Online Resources: Sl.No Website Link 1 https://www.electronicshub.org 2 https://www.elprocus.com 3 https://www.tutorialspoint.com

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Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding sensor interfacing with 8051. Stepper motor – operation, explain the interfacing with 8051 DC motor – operation, explain the interfacing with 8051 4x4 -Matrix keyboard – Structure, identifying key press, explain the interfacing with 8051. 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051. Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor. Text / Reference: T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller T1 and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Subratha Ghoshal – 8051 microcontroller internals, instructions, programming T2 and interfacing- Pearson T3 Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition. Online Resources: Sl.No Website Link 1 https://www.electronicshub.org 2 https://www.elprocus.com 3 https://www.tutorialspoint.com using standard technical terminology.
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Engineering or professional relevance

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8051 - with adc, with dac, with temperature control system using lm35 temperature sensor. text / reference: t/r book title/author muhammad ali mazidi and janice gillispie mazidi - the 8051 microcontroller t1 and embedded systems using assembly and c - pearson education- second edition. subratha ghoshal – 8051 microcontroller internals, instructions, programming t2 and interfacing- pearson t3 kenneth j ayala - the 8051 microcontroller - thomson - third edition. online resources: sl.no website link 1 https://www.electronicshub.org 2 https://www.elprocus.com 3 https://www.tutorialspoint.com should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding sensor interfacing with 8051. Stepper motor – operation, explain the interfacing with 8051 DC motor – operation, explain the interfacing with 8051 4x4 -Matrix keyboard – Structure, identifying key press, explain the interfacing with 8051. 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051. Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor. Text / Reference: T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller T1 and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Subratha Ghoshal – 8051 microcontroller internals, instructions, programming T2 and interfacing- Pearson T3 Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition. Online Resources: Sl.No Website Link 1 https://www.electronicshub.org 2 https://www.elprocus.com 3 https://www.tutorialspoint.com means in the official course context
Study lens	Principle → components or stages → result → application
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Malayalam support: Module 4-□□ 4 Understanding sensor interfacing with 8051. Stepper motor – operation, explain the interfacing with 8051 DC motor – operation, explain the interfacing with 8051 4x4 -Matrix keyboard – Structure, identifying key press, explain the interfacing with 8051. 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051. Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor. Text / Reference: T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller T1 and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Subratha Ghoshal – 8051 microcontroller internals, instructions, programming T2 and interfacing- Pearson T3 Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition. Online Resources: Sl.No Website Link 1 https://www.electronicshub.org 2 https://www.elprocus.com 3 https://www.tutorialspoint.com □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain Compare Microprocessor and Microcontroller. (3 marks)**

▶ **Define or explain List the features of 8051 microcontroller. 1 Remembering Explain the block diagram of 8051. (3 marks)**

▶ **Define or explain 2 Understanding microcontroller.. (3 marks)**

▶ **Define or explain Explain the memory of 8051. 7 Understanding Features and Block diagram of 8051 Introduction to microprocessors and microcontrollers, features of 8051, simple block diagram of 8051, Data memory organization, program memory organization, PSW register.. (3 marks)**

Module 2

▶ **Define or explain 3 Understanding Programming. Explain various addressing modes and types. (3 marks)**

▶ **Define or explain 4 Understanding of instructions. Describe Jump, Loop, Call and single bit level. (3 marks)**

▶ **Define or explain 4 Understanding instructions. Develop Assembly language programs for. (3 marks)**

▶ **Define or explain 6 Applying 8051. Introduction to Assembly Language programming of 8051 – Structure, addressing modes, Types of instructions: -Data transfer, arithmetic, logical, Compare, Rotate and swap instructions. Describe jump instruction- conditional and unconditional, call instructions, calling subroutines. machine cycle, Delay generation. Single-Bit level instructions, Data serialization. Simple programs: - 8-bit Addition, 8-bit subtraction, multiplication, division, ASCII to packed BCD, packed BCD to ASCII, BCD to hex and hex to BCD, port reading and writing.. (3 marks)**

Module 3

▶ **Define or explain 4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating. (3 marks)**

▶ **Define or explain 4 Understanding modes. Illustrate TCON register and time delay. (3 marks)**

▶ **Define or explain 3 Understanding calculation for Timer. Illustrate special function registers for serial. (3 marks)**

▶ **Define or explain 3 Understanding communication. Interrupts, Timers, Serial communication. 8051 Interrupts: Concept, Types, Interrupt service routine (ISR), IE register, Steps in executing an interrupt, Interrupt priority, IP register. Timer: TMOD register, Timer operating modes, TCON register, Time Delay calculation. Basics of serial communication, Baud rate, 8051 connections to RS232 connector, SBUF register, SCON register and Different serial data transmission modes in 8051. PCON register.. (3 marks)**

Module 4

▶ **Define or explain 4 Understanding 8051. Explain the interfacing of DC motor for speed. (3 marks)**

▶ **Define or explain 3 Understanding control. Illustrate the interfacing of 4x4 keyboard and. (3 marks)**

▶ **Define or explain 4 Understanding 16x2 LCD system with 8051. Illustrate ADC, DAC and LM35 temperature. (3 marks)**

▶ **Define or explain 4 Understanding sensor interfacing with 8051. Stepper motor – operation, explain the interfacing with 8051 DC motor – operation, explain the interfacing with 8051 4x4 -Matrix keyboard – Structure, identifying key press, explain the interfacing with 8051. 16x2 LCD - Pin description, advantages of LCD, explain the interfacing with 8051. Explain the interfacing of 8051 - with ADC, with DAC, with temperature control system using LM35 temperature sensor. Text / Reference: T/R Book Title/Author Muhammad Ali Mazidi and Janice Gillispie Mazidi - The 8051 Microcontroller T1 and Embedded Systems Using Assembly and C - Pearson Education- Second Edition. Subratha Ghoshal – 8051 microcontroller internals, instructions, programming T2 and interfacing- Pearson T3 Kenneth J Ayala - The 8051 Microcontroller - Thomson - Third Edition. Online Resources: Sl.No Website Link 1 https://www.electronicshub.org 2 https://www.elprocus.com 3 https://www.tutorialspoint.com. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Compare Microprocessor and Microcontroller**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Understanding Programming. Explain various addressing modes and types**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **4 Understanding Interrupts and handling of interrupts. Explain TMOD register and timer operating**.

Module 4

1-mark definition: State the meaning of **4 Understanding 8051. Explain the interfacing of DC motor for speed**.

3-mark explanation: Explain one principle, process or comparison from

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://www.electronicshub.org https://www.elprocus.com https://www.tutorialspoint.com

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 5492. Verify institutional updates against the current official curriculum.